

UNIVERSITY OF BRISTOL

Faculty of Science

MOCK EXAMINATION — Paper 7 (Hard)**Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Duration: 3 Hours

Total Marks: 100

Instructions:

- Answer **ALL** questions.
 - **Section A** (Oscillations and Waves): 10 multiple-choice questions, 20 marks total.
 - **Section B** (Quantum Physics): 10 multiple-choice questions, 20 marks total.
 - **Section C**: 4 long-answer questions, 60 marks total.
 - An approved calculator is permitted.
 - A formula sheet is provided at the end of this paper.
 - Show all working for Section C. Marks will be awarded for clear, logical solutions.
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Section A: Oscillations and Waves

Answer all 10 questions. Each question is worth 2 marks.

1. A particle of mass m moves in a potential $V(x) = V_0(x/a)^2$, where V_0 and a are positive constants. The particle undergoes small oscillations about the equilibrium at $x = 0$. What is the angular frequency ω_0 of these oscillations? [2 marks]

A. $\sqrt{\frac{V_0}{ma^2}}$

B. $\sqrt{\frac{2V_0}{ma^2}}$

C. $\sqrt{\frac{V_0}{2ma^2}}$

D. $\sqrt{\frac{4V_0}{ma^2}}$

2. A damped harmonic oscillator has damping constant γ . The amplitude decays as $A(t) = A_0 e^{-\gamma t/2}$. After a time $t = 2/\gamma$, what fraction of the *initial energy* remains? [2 marks]

A. $1/e$

B. $1/e^2$

C. $1/e^4$

D. $1/\sqrt{e}$

3. A forced oscillator has natural frequency ω_0 and damping constant γ . The phase lag between the driving force and the steady-state response is given by $\tan \delta = \gamma\omega/(\omega_0^2 - \omega^2)$. At what driving frequency ω is the phase lag exactly $\pi/4$? [2 marks]

A. $\omega = \omega_0$

B. $\omega = \omega_0 - \gamma$

C. $\omega = -\frac{\gamma}{2} + \sqrt{\omega_0^2 + \frac{\gamma^2}{4}}$

D. $\omega = \frac{\gamma}{2} + \sqrt{\omega_0^2 + \frac{\gamma^2}{4}}$

4. A transverse wave travels from a string with impedance Z_1 to a string with impedance $Z_2 = 3Z_1$. What fraction of the incident power is transmitted? [2 marks]

A. $1/4$

B. $3/4$

C. 9/16

D. 3/16

5. Two identical coupled oscillators have normal mode frequencies ω_1 and ω_2 (with $\omega_2 > \omega_1$). When one oscillator is displaced and released, energy transfers back and forth between them. What is the beat (energy-exchange) frequency? [2 marks]

A. $\omega_1 + \omega_2$

B. $\omega_1\omega_2$

C. $(\omega_2 - \omega_1)/2$

D. $\omega_2 - \omega_1$

6. An organ pipe of length L is open at one end and closed at the other. The 2nd overtone of this pipe corresponds to which harmonic? [2 marks]

A. The 3rd harmonic ($n = 3$)

B. The 4th harmonic ($n = 4$)

C. The 5th harmonic ($n = 5$)

D. The 6th harmonic ($n = 6$)

7. A sound source produces a sound level of 83 dB at a certain point. If the intensity is reduced by a factor of 100, what is the new sound level at that point? [2 marks]

A. 63 dB

B. 0.83 dB

C. 81 dB

D. 41.5 dB

8. A sound source emitting frequency f and an observer are both moving in the *same* direction. The source moves at speed v_s and the observer at speed v_o , with $v_s > v_o$ (source faster, approaching the observer from behind). Taking v as the speed of sound, which expression gives the frequency f' heard by the observer? [2 marks]

A. $f' = f \frac{v + v_o}{v + v_s}$

B. $f' = f \frac{v - v_o}{v - v_s}$

C. $f' = f \frac{v + v_o}{v - v_s}$

D. $f' = f \frac{v - v_o}{v + v_s}$

9. A medium has the dispersion relation $\omega^2 = \omega_0^2 + c^2 k^2$, where ω_0 and c are constants. What is the group velocity v_g ? [2 marks]

- A. c
- B. ω/k
- C. $\frac{c^2 k}{\sqrt{\omega_0^2 + c^2 k^2}}$
- D. $\frac{\omega_0^2 + c^2 k^2}{ck}$

10. N identical sound sources, each of intensity I_0 , are placed at the same location but with *random* (uncorrelated) phases. How does the total intensity scale with N ? [2 marks]

- A. $I_{\text{total}} = N^2 I_0$
- B. $I_{\text{total}} = N I_0$
- C. $I_{\text{total}} = \sqrt{N} I_0$
- D. $I_{\text{total}} = I_0$ (independent of N)

Section B: Quantum Physics

Answer all 10 questions. Each question is worth 2 marks.

1. For two normalised quantum states $|\psi\rangle$ and $|\phi\rangle$, what is $\text{Tr}(|\psi\rangle\langle\phi|)$? [2 marks]

- A. $\langle\psi|\phi\rangle$
- B. $\langle\phi|\psi\rangle$
- C. $|\langle\phi|\psi\rangle|^2$
- D. 1

2. The commutator $[\sigma_x, \sigma_z] = \sigma_x\sigma_z - \sigma_z\sigma_x$ is equal to: [2 marks]

- A. $2i\sigma_y$
- B. $-2i\sigma_y$
- C. $i\sigma_y$
- D. 0

3. A spin- $\frac{1}{2}$ particle is in a state for which $\langle S_x \rangle = +\hbar/2$ (the maximum possible value). What can you conclude about $\langle S_z \rangle$ in this state? [2 marks]

- A. $\langle S_z \rangle = +\hbar/2$
- B. $\langle S_z \rangle = -\hbar/2$
- C. $\langle S_z \rangle = 0$
- D. $\langle S_z \rangle$ cannot be determined without more information

4. A Hermitian operator $\hat{A}$ has a degenerate eigenvalue a with a two-dimensional eigenspace. A system is in a state $|\psi\rangle$ that is *not* an eigenstate of $\hat{A}$, but has a non-zero overlap with the eigenspace. After a measurement of $\hat{A}$ yields the result a , the post-measurement state is: [2 marks]

- A. Any one of the degenerate eigenstates, chosen at random.
- B. The projection of $|\psi\rangle$ onto the degenerate eigenspace, normalised.
- C. Unchanged — the state remains $|\psi\rangle$.
- D. An equal superposition of all eigenstates of $\hat{A}$.

5. The expectation value $\langle \hat{A} \rangle$ of an observable $\hat{A}$ in a state $|\psi(t)\rangle$ evolving under a Hamiltonian H is time-independent if: [2 marks]

- A. $\hat{A}$ is Hermitian.

- B. $\hat{A}$ commutes with H and $\hat{A}$ has no explicit time dependence.
- C. $\hat{A}$ is unitary.
- D. $|\psi\rangle$ is normalised.
6. A spin- $\frac{1}{2}$ particle is prepared in the state $|\uparrow_{\hat{n}}\rangle$ where $\hat{n}$ is a unit vector making an angle θ with the z -axis. What is the probability of measuring $S_z = +\hbar/2$? [2 marks]
- A. $\cos \theta$
- B. $\sin^2(\theta/2)$
- C. $\cos^2(\theta/2)$
- D. $\frac{1}{2}(1 + \sin \theta)$
7. Consider a composite system of two particles: particle 1 has spin $s_1 = 1$ and particle 2 has spin $s_2 = \frac{1}{2}$. What is the dimension of the composite Hilbert space? [2 marks]
- A. 3
- B. 4
- C. 5
- D. 6
8. Given a bipartite system of particles 1 and 2, can you determine whether the two-particle state is entangled by performing measurements *only* on particle 1? [2 marks]
- A. Yes — if the outcomes are random, the state is entangled.
- B. Yes — if the expectation values differ from $\pm\hbar/2$, the state is entangled.
- C. No — the reduced state of particle 1 alone cannot distinguish entangled from mixed states.
- D. No — entanglement is not a physical property and cannot be measured.
9. Three identical fermions are to be placed in 5 distinct quantum states. How many valid configurations are there (respecting the Pauli exclusion principle)? [2 marks]
- A. 5
- B. 10
- C. 35
- D. 60
10. Which of the following quantities is *not* conserved in weak interactions? [2 marks]
- A. Electric charge

- B. Baryon number
- C. Quark flavour (e.g. strangeness)
- D. Lepton number

Section C: Long Answer Questions

Answer all 4 questions. Show all working.

Question 1.

[15 marks]

Rolling, Angular Momentum, and Reduced Mass

- (a) A solid sphere of mass $M = 3$ kg and radius $R = 0.1$ m rolls without slipping on a horizontal surface at speed $v = 5$ m/s. It then encounters an inclined plane at angle $\theta = 30^\circ$ and rolls up it (without slipping).
- (i) Find the total kinetic energy (translational plus rotational) of the sphere on the flat surface. 2
 - (ii) Find the distance along the incline the sphere travels before momentarily coming to rest. 2
- (b) A uniform disk of mass $M = 2$ kg and radius $R = 0.3$ m is free to rotate about a fixed axis through its centre. A string wound around the rim is pulled with a constant force $F = 10$ N.
- (i) Find the moment of inertia of the disk and the angular acceleration produced by the applied force. 2
 - (ii) Starting from rest, find the angular velocity after 4 s and the rotational kinetic energy at that instant. 2
- (c) An ice skater spinning with arms extended has moment of inertia $I_1 = 4$ kg $\cdot$ m² and angular velocity $\omega_1 = 2$ rev/s. She pulls her arms in, reducing her moment of inertia to $I_2 = 1.5$ kg $\cdot$ m². Find her new angular velocity. Calculate the ratio of final to initial rotational kinetic energy. Explain where the extra energy comes from. 3
- (d) Two objects with masses $m_1 = 3$ kg and $m_2 = 1$ kg are connected by a spring of spring constant $k = 48$ N/m and placed on a frictionless surface. Using the concept of reduced mass, find the frequency of oscillation of this system. If the spring is initially compressed by 0.2 m and the system is released from rest, find the maximum speed of each mass in the centre-of-mass frame. 4

Question 2.

[15 marks]

Multi-Step Thermodynamic Cycle with Entropy Analysis

A monatomic ideal gas ($n = 1$ mol, $\gamma = 5/3$) undergoes the following cycle:

State A: $P_A = 1.0 \times 10^5$ Pa, $V_A = 0.025$ m³.

$A \rightarrow B$: Isobaric expansion to $V_B = 0.075$ m³.

$B \rightarrow C$: Isochoric (constant volume) cooling to $P_C = P_A/3$.

$C \rightarrow A$: A process that returns the gas to state A.

- (a) Find the temperatures T_A , T_B , and T_C . Calculate Q , W , and ΔU for each of the steps $A \rightarrow B$ and $B \rightarrow C$. 4
- (b) Determine the nature of the process $C \rightarrow A$ (is it isothermal, adiabatic, or neither?). Calculate W , Q , and ΔU for step $C \rightarrow A$. Verify that $\Delta U_{\text{cycle}} = 0$. 4
- (c) Calculate the efficiency of this cycle. Compare it to the Carnot efficiency operating between the highest and lowest temperatures in the cycle. 3
- (d) Calculate the entropy change of the gas for each step of the cycle. Verify that the total entropy change of the gas over one complete cycle is zero. Assuming each heat exchange occurs with a reservoir at the appropriate extreme temperature (T_B for heating, T_C for cooling), calculate the total entropy change of the universe for one cycle and verify that it is positive. 4

Question 3.

[15 marks]

Wave Equation Derivation, Power, and Dispersion

- (a) Starting from Newton's second law applied to a small element of a stretched string under tension T with linear mass density μ , derive the wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

and show that $v = \sqrt{T/\mu}$. 4

- (b) Show that the average power transmitted by a harmonic wave $y = A \sin(kx - \omega t)$ on a string is $\langle P \rangle = \frac{1}{2} \mu \omega^2 A^2 v$.

Hint: The power is given by the product of the transverse force $F_y = -T \partial y / \partial x$ and the transverse velocity $\partial y / \partial t$ at a point on the string. 4

- (c) A string of linear mass density $\mu_1 = 0.01$ kg/m is joined to a string with $\mu_2 = 0.04$ kg/m. Both are under tension $T = 100$ N. A harmonic wave with amplitude $A_i = 5$ mm and frequency $f = 200$ Hz travels on string 1 toward the junction. Calculate the power carried by the incident wave, the reflected wave, and the transmitted wave. 3

- (d) The dispersion relation for waves on a loaded string is $\omega(k) = 2\omega_0 \left| \sin\left(\frac{ka}{2}\right) \right|$, where a is the spacing between masses and ω_0 is a constant. Find the phase velocity $v_p(k)$ and the group velocity $v_g(k)$. At what value of k are v_p and v_g equal? What happens to v_g at $k = \pi/a$? 4

Question 4.

[15 marks]

Bloch Sphere, Larmor Precession, and Entanglement

- (a) A spin- $\frac{1}{2}$ particle is in the general state

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |\uparrow_z\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |\downarrow_z\rangle,$$

which describes a spin pointing in the direction (θ, ϕ) on the Bloch sphere.

- (i) Verify that this state is normalised. 1
 - (ii) Calculate $\langle S_z \rangle$, $\langle S_x \rangle$, and $\langle S_y \rangle$ in terms of θ and ϕ . 3
- (b) The Hamiltonian of a spin- $\frac{1}{2}$ particle in a magnetic field along the x -direction is
- $$H = \omega_0 S_x = \frac{\omega_0 \hbar}{2} \sigma_x.$$
- (i) Find the energy eigenvalues and eigenstates of this Hamiltonian. 2
 - (ii) If the particle starts in the state $|\uparrow_z\rangle$ at $t = 0$, find $|\psi(t)\rangle$. Express your answer in the z -basis. 2
- (c) For the state found in part (b)(ii), calculate the probabilities $P(\uparrow_z, t)$ and $P(\downarrow_z, t)$. Interpret the result physically — what is the spin doing? 3
- (d) Consider two spin- $\frac{1}{2}$ particles in the singlet state

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

- (i) Show that if S_z is measured on particle 1, the reduced density matrix of particle 2 is $\rho_2 = \frac{1}{2} \hat{I}$ (the maximally mixed state), regardless of the measurement outcome. 2
- (ii) Is this property unique to S_z measurements, or does it also hold if S_x is measured on particle 1? Justify your answer. 2

Formula Sheet

Mathematical Formulae

Binomial expansion: $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (|x| \ll 1)$

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

Trig identities: $\sin A + \sin B = 2 \cos\left(\frac{A-B}{2}\right) \sin\left(\frac{A+B}{2}\right)$
 $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

Mechanics

SUVAT: $v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u+v)t$

Dot product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta = a_x b_x + a_y b_y + a_z b_z$

Cross product: $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$

Rocket equation: $\Delta v = u \ln\left(\frac{m_0}{m_f}\right)$

Gravitational PE: $U = -\frac{GMm}{r}$

Escape velocity: $v_e = \sqrt{\frac{2GM}{R}}$

Force from PE: $F = -\frac{dU}{dx}$

Elastic collision (1D): $v_1' = \frac{m_1 - m_2}{m_1 + m_2}v_1 + \frac{2m_2}{m_1 + m_2}v_2$

Centre of mass: $\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}$

Reduced mass: $\mu = \frac{m_1 m_2}{m_1 + m_2}$

Centripetal accel: $a_c = \frac{v^2}{r} = \omega^2 r$

Parallel axis thm: $I = I_{\text{cm}} + Md^2$

Atwood machine: $a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$

Moments of inertia (about axis through centre of mass):

Thin ring (radius R):	$I = MR^2$
Solid disk (radius R):	$I = \frac{1}{2}MR^2$
Solid sphere (radius R):	$I = \frac{2}{5}MR^2$
Hollow sphere (radius R):	$I = \frac{2}{3}MR^2$
Thin rod (length L , about centre):	$I = \frac{1}{12}ML^2$
Thin rod (length L , about end):	$I = \frac{1}{3}ML^2$

Properties of Matter

Ideal gas: $PV = nRT$

Heat capacities: $C_P - C_V = R$ (per mole)

Adiabatic: $PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$

Isothermal work: $W = nRT \ln\left(\frac{V_2}{V_1}\right)$

Carnot efficiency: $\eta = 1 - \frac{T_c}{T_h}$

Entropy: $dS = \frac{dQ_{\text{rev}}}{T}, \quad \Delta S = mc \ln\left(\frac{T_2}{T_1}\right), \quad \Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$

Lennard-Jones: $V(r) = 4\varepsilon \left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^6 \right]$

Mean KE: $\langle KE \rangle = \frac{3}{2}k_B T$

MB speeds: $c_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}, \quad c_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}, \quad c_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$

Oscillations and Waves

SHM:	$x(t) = A \cos(\omega_0 t + \phi), \quad \omega_0 = \sqrt{k/m}$
Damped SHM:	$x(t) = A e^{-\gamma t/2} \cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$
Quality factor:	$Q = \omega_0/\gamma$
Forced amplitude:	$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}}$
Phase lag:	$\tan \delta = \frac{\gamma \omega}{\omega_0^2 - \omega^2}$
Pendulums:	$T_{\text{simple}} = 2\pi \sqrt{L/g}, \quad T_{\text{physical}} = 2\pi \sqrt{I/(Mgh)}$
Wave equation:	$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$
String speed:	$v = \sqrt{T/\mu}$
Sound in gas:	$v = \sqrt{\gamma P/\rho}$
Harmonic wave:	$y = A \sin(kx - \omega t)$
String impedance:	$Z = \mu v = \sqrt{\mu T}$
Reflection:	$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{2Z_1}{Z_1 + Z_2}$
Power (string):	$\langle P \rangle = \frac{1}{2} \mu \omega^2 A^2 v$
Decibels:	$\beta = 10 \log_{10}(I/I_0), \quad I_0 = 10^{-12} \text{ W/m}^2$
Doppler:	$f' = f \frac{v \pm v_o}{v \mp v_s}$
Group velocity:	$v_g = \frac{d\omega}{dk}$
String harmonics:	$f_n = \frac{nv}{2L} \quad (n = 1, 2, 3, \dots)$
Open-closed pipe:	$f_n = \frac{nv}{4L} \quad (n = 1, 3, 5, \dots)$

Quantum Physics

Schrödinger eq: $i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$

Time evolution: $|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle, \quad U(t) = e^{-iHt/\hbar}$

Spin operators:

$$S_z = \frac{\hbar}{2} (\uparrow\uparrow\langle\uparrow - \downarrow\downarrow\langle\downarrow)$$

$$S_x = \frac{\hbar}{2} (\uparrow\uparrow\langle\downarrow + \downarrow\downarrow\langle\uparrow)$$

$$S_y = \frac{\hbar}{2} (-i \uparrow\uparrow\langle\downarrow + i \downarrow\downarrow\langle\uparrow)$$

Spin eigenstates:

$$\uparrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + \downarrow_z\rangle), \quad \downarrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - \downarrow_z\rangle)$$

$$\uparrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + i \downarrow_z\rangle), \quad \downarrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - i \downarrow_z\rangle)$$

Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Physical Constants

Speed of light	c	3.00×10^8 m/s
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck's constant	$\hbar$	1.055×10^{-34} J·s
Boltzmann constant	k_B	1.381×10^{-23} J/K
Gas constant	R	8.314 J/(mol·K)
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
Gravitational constant	G	6.674×10^{-11} N·m ² /kg ²
Acceleration due to gravity	g	9.81 m/s ²
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Elementary charge	e	1.602×10^{-19} C
Mass of Earth	$M_\oplus$	5.97×10^{24} kg
Radius of Earth	$R_\oplus$	6.37×10^6 m